Microsoft Elevate

CAPSTONE PROJECT

Credit Card Financial Dashboard using Power BI

PRESENTED BY

STUDENT NAME: ARIJIT MODAK

COLLEGE NAME: MAKAUT UNIVERSITY

DEPARTMENT: BCA

EMAIL ID: arijitmodak006@gmail.com

PHONE NO: 7047735629

OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
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PROBLEM STATEMENT:

Financial institutions generate a large volume of credit card transaction data. However, without proper visualization and analysis, it becomes difficult to identify revenue trends, customer segments, spending behavior, and interest income.
The challenge is to analyze the 2023 credit card dataset and generate meaningful business insights using an interactive dashboard.

PROPOSED SOLUTION:

To solve this problem, an interactive Power BI dashboard has been developed to analyze credit card financial data .The dashboard provides insights into revenue, transaction amount, customer demographics, interest earned, and spending patterns.

Key Components:

- Data cleaning and transformation using Power Query
- Data modeling and relationship creation
- DAX measures for KPIs
- Interactive visualizations
- Slicers for dynamic filtering (Quarter, Gender, Income group, Card type)

SYSTEM APPROACH:

Tools Used:

- Microsoft Power BI
- SQL (for data querying)
- CSV Dataset Files
- DAX (Data Analysis Expressions)

Development Steps:

1. Import dataset into Power BI
2. Data cleaning & transformation
3. Create relationships between tables
4. Create calculated columns and measures
5. Design interactive dashboard visuals
6. Validate insights with dataset

ALGORITHM & DEPLOYMENT:

Data Processing Algorithm:

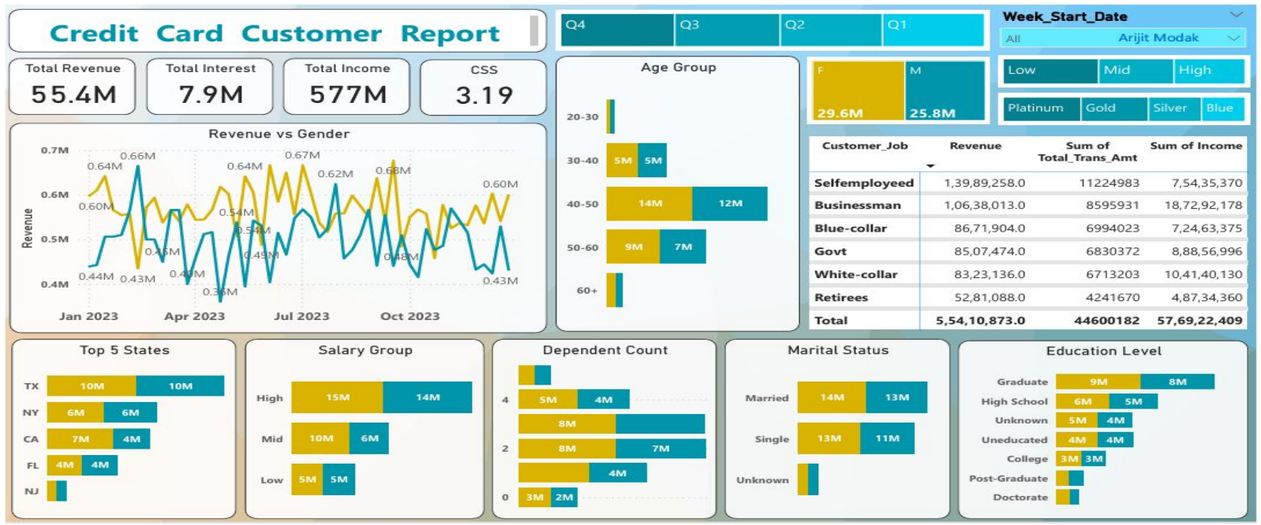
- Data Filtering
- Aggregation (SUM, COUNT, AVERAGE)
- KPI Calculation (Revenue, Interest, Income)
- Segmentation by Age, Gender, Salary Group

Deployment:

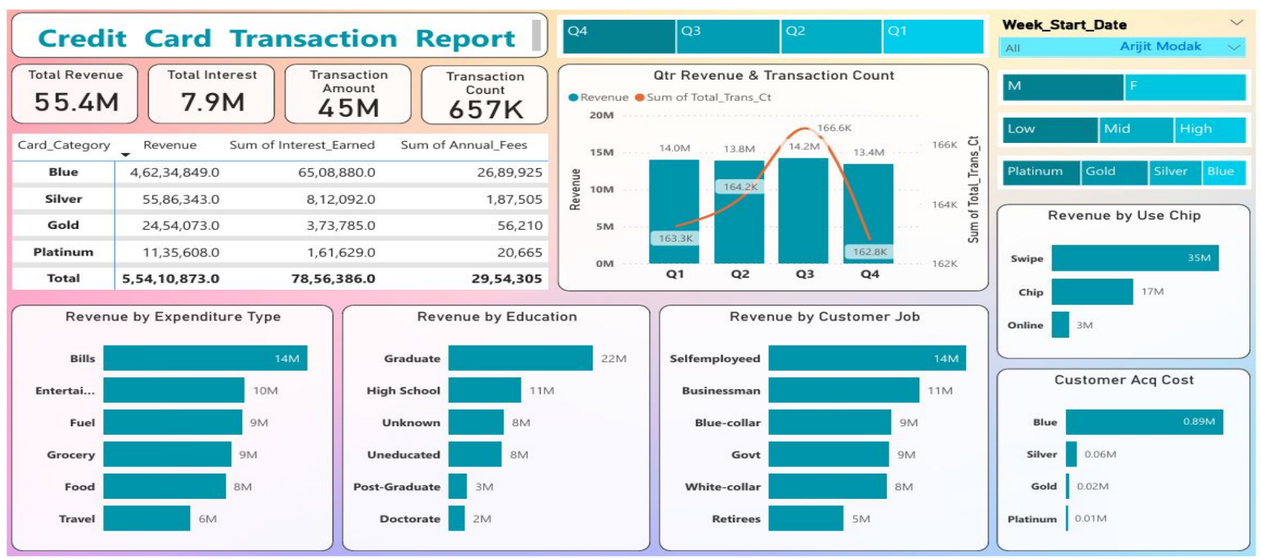
- Dashboard exported as PDF
- Project uploaded to GitHub repository
- Report submitted for internship evaluation

RESULT:

Result 1: Customer Analysis Dashboard



Result 2: Transaction & Revenue Dashboard



Customer Analysis Dashboard:
The dashboard provides insights into customer demographics, income groups, job roles, and gender distribution. It helps identify high-value customers and analyze spending patterns.
Dashboard Link:
Drive – https://drive.google.com/file/d/16VccXF2yii1WS8KAnhFKZNT9cFP6qrl3/view?usp=drivesdk

Transaction & Revenue Dashboard:
This dashboard analyzes total revenue, transaction amount, interest earned, and quarterly performance trends. It helps understand financial growth and card category performance.
Dashboard Link:
Drive – https://drive.google.com/file/d/1kqjShW6AXpZN0gCt5cVB5khkJFPzgTpw/view?usp=drivesdk

CONCLUSION:

The Credit Card Financial Dashboard effectively analyzes financial data for the year 2023.
It helps understand customer behavior, revenue trends, and financial performance.
The project demonstrates the practical use of Power BI for business intelligence and decision-making support.

FUTURE SCOPE:

- The dashboard can be integrated with **real-time financial data** for live insights.
- **Machine learning models** can be used to predict customer spending and credit risk.
- More datasets like **loan data and credit scores** can be added for better analysis.
- The project can be converted into an **interactive web-based dashboard** for wider access.

REFERENCES:

GitHub Repository :

Project Source Code :
https://github.com/ArijitModak06CODE/Credit-Card-Financial-Dashboard-

Dataset :

The following datasets were used in this project :

- **customer.csv** – Customer information data
  Drive link- https://drive.google.com/file/d/1a84iTEPFdpBmiOH9zUIc6hTtMbPL_Hj-/view?usp=drivesdk
- **cust_add.csv** – Additional customer details
  Drive link- https://drive.google.com/file/d/1Muo4ncGzwuhF2hvLpFLejM7Df9MLeRee/view?usp=drivesdk
- **credit_card.csv** – Credit card transaction data
  Drive link- https://drive.google.com/file/d/19WsPgBPYmAsG7Hy7jsEzReyn0bCdoRr6/view?usp=drivesdk
- **cc_add.csv** – Additional credit card attributes
  Drive link- https://drive.google.com/file/d/1Ltt81BcswXtxxuZ-b8Sq8K0ZmqMyZrz9/view?usp=drivesdk

Tools Documentation :

- Microsoft Power BI Documentation
- DAX Documentation

Thank You